Two Axes That Cross — Trust Against Placement

Who a tenant is, what sits inside one, and in which cells that tenant is allowed to exist. The NATS trust hierarchy names **who** and carries no geography; the placement hierarchy names **where** and carries no identity. They cross; neither one contains the other. **This is a directional document:** a first-pass position recorded so the platform can move, not a settled decision, stated as at 17 September 2026. Nothing on this sheet is built — the lab runs one server in one region. Subjects, streams and grants belong to LB-V3-L3-05; residency and border rules belong to LB-V3-L3-08; claim text and schemas belong to L4.

TRUST HIERARCHY · WHO — THERE IS NO GEOGRAPHY ON THIS AXIS

- **O Operator** — the one trust root
  signs account claims, never a user
  proposed · signing keys held in ZA
- **T Tenant account** — the hard messaging wall
  one per tenant · a NATS account
  proposed · no V3 ADR
- **G Organisation, membership** — roles and attributes, inside
  an org never makes a new account
  open · architecture to confirm
- **U User credential** — short-lived, issued by its account
  a person session, or a workload
  proposed · no V3 ADR

PLACEMENT HIERARCHY · WHERE — THERE IS NO IDENTITY ON THIS AXIS

- **R Region** — a place a cell is deployed in
  South Africa, Australia, Botswana
  proposed · no V3 ADR
- **C Cell** — one region's complete runtime
  ingress, web, NATS, workflow, stores
  proposed · no V3 ADR
- **D JetStream domain** — marks one cell
  a domain is a place, not a tenant
  proposed · see LB-V3-L3-05
- **S Cell-resident runtime** — every replica and every store
  nothing here names a tenant
  proposed · no V3 ADR

the trust axis carries no region

the placement axis carries no tenant

WHERE THEY CROSS · AN ACCOUNT IS A ROW, A CELL IS A COLUMN

no regulated business path crosses this line

accounts, down / cells, across	ZA shared cell — Cape Town · ZAR default	AU shared cell — Sydney · AUD default	BW sovereign cell — open · legal
$SYS — standing account	**Administration** — server and JetStream admin only	**Administration** — server and JetStream admin only	**Administration** — server and JetStream admin only
CONTROL — standing account	**Control plane** — NATS provisioning, identity, observability	**Control plane** — NATS provisioning, identity, observability	**Control plane** — NATS provisioning, identity, observability
PLATFORM — standing account	**Shared data** — reference data and registry, imported	**Shared data** — reference data and registry, imported	**Shared data** — reference data and registry, imported
TENANT A — two permitted cells	**Streams, KV, Postgres, files** — ZA business records, owned here	**Streams, KV, Postgres, files** — AU business records, owned here	**Not permitted** — the placement record refuses this cell
TENANT B — one permitted cell	**Streams, KV, Postgres, files** — ZA business records, owned here	**Not permitted** — the placement record refuses this cell	**Not permitted** — the placement record refuses this cell
TENANT BW — sovereign only	**Not permitted** — the placement record refuses this cell	**Not permitted** — the placement record refuses this cell	**Streams, KV, Postgres, files** — BW records · nothing regulated leaves

! The account carries the tenant · the cell carries the region
two axes that cross — neither one contains the other
a picture that nests a tenant inside a cell is wrong
as at 2026-09-17, CONTROL row added 2026-10-19 · no V3 ADR

X Nothing regulated crosses the sovereign edge
no gateway, no mirror, no shared account band
only signed policy and account claims enter, and they carry no payload
open · legal confirmation required before this is compliance evidence

Read the top as two separate trees, then read the grid as a crossing. A row is one NATS account and it spans every cell it is permitted in. A column is one cell and it holds many accounts. A red square is the placement record refusing, not a hole in the drawing. **Built today:** nothing on this sheet — the lab is one server in one region. **Not shown:** subjects, streams and grants, which are `LB-V3-L3-05`; residency and border rules, which are `LB-V3-L3-08`; claim text and schemas, which are `L4`.

SCOPE	**One question.** Who is a tenant, and in which cells may that tenant exist? What runs inside one cell is sheet 2.
DIRECTION	**Directional.** A first pass as at 2026-09-17 so work can start. Not a decision, and not compliance evidence.
AXES	**Two hierarchies.** Trust names **who**. Placement names **where**. Neither one contains the other.
ACCOUNT	A tenant is **one account**. It keeps one identity in every cell it is permitted to use.
CELL	A **cell** is one region's complete runtime. It is a place, and it names no tenant.
ORG	Organisations live **inside** an account. A role or a membership never makes a second NATS account.
DOMAIN	A JetStream **domain** marks a cell. It is not a tenancy boundary; the account is.
GRID	Read a **row** for an account and a **column** for a cell. A red square is a refusal, not a gap.
STANDING	**$SYS** administers servers only. **CONTROL** holds the control plane. **PLATFORM** holds shared data that tenants import.
OPEN	Every amber tile needs a named owner. Seven open items are listed in the **L3-01 register**.

LEGEND

- **Green boundary** — the trust hierarchy: who
- **Dashed amber tile** — open, a named owner must confirm
- **Green line** — a trust fact carried into the grid
- **Line label** — names the relationship, never bare "uses"
- **Cyan boundary** — the placement hierarchy: where
- **Red square or panel** — a combination that is refused
- **Cyan dotted line** — a placement fact carried in
- **Amber boundary** — where the two axes cross
- **Amber panel** — a rule that governs the whole sheet
- **Red barred line** — a border nothing regulated crosses
- **Dashed grey tile** — proposed for V3, not built
- **Solid tile** — measured in the lab; there is none here
- **Arrow head** — the direction of the relationship

WORDS ON THIS SHEET

- **operator** the one NATS signing root
- **account** the NATS tenancy boundary
- **user** a short-lived NATS credential
- **organisation** a business unit inside a tenant
- **membership** a person's place in an organisation
- **tenant** a marketplace-operating business
- **cell** one region's complete runtime
- **region** a place a cell is deployed in
- **domain** a JetStream name marking one cell
- **placement record** names a tenant's permitted cells
- **$SYS** the NATS system account
- **CONTROL** the control-plane account
- **PLATFORM** the shared data account
- **sovereign** kept inside one country by law
- **KV** NATS key-value, a fast lookup store
- **ADR** architecture decision record

Traceability: L3-01-001 to L3-01-018, L3-01-032 to L3-01-033 and L3-01-049 to L3-01-050 (trust hierarchy, standing accounts, placement hierarchy, the crossing grid, refusals, directional marking). Parent: LB-V3-L2-01, its NATS trust group on sheet 1 and its regional cell model on sheet 2. Related: LB-V3-L3-05 Messaging / NATS; LB-V3-L3-08 Multi-Region.

Inside One Square — Which Services Run Where, and Whose Credential They Hold

One cell on sheet 1 is one square in the grid. This sheet opens that square. The cell owns the runtime and names no tenant. A service replica runs in the cell and **borrows** a short-lived user credential to act for one tenant account. Every durable record that replica writes lands inside that account. **This is a directional document** as at 17 September 2026, except the bottom band, which is measured in the demo 03 lab. Service responsibilities belong to LB-V3-L2-01; subjects and streams belong to LB-V3-L3-05; packaging and scaling belong to LB-V3-L3-12.

THE CELL OWNS THESE · ONE OF EACH PER CELL, AND NO TENANT IS NAMED

I — Regional ingress and web
the way in, served locally
`WAF, API edge, shell + MFE registry`
`proposed · no V3 ADR`

N — NATS, three servers
plus a local claim resolver
`a logical HA minimum, not sizing`
`proposed · sizing is L4`

W — Workflow runtime
long-running work, cell-resident
`Temporal workers and persistence`
`proposed · no V3 ADR`

O — Stores and operations
the cell's own machinery
`Postgres per owner, evidence adapter`
`masked logs, key handling`

the cell hosts every replica, and names no tenant while doing it

THE SERVICE REPLICAS · SHARED IN THE CELL, AND THEY BORROW A USER

C — Control-plane slice
tenant lifecycle, identity, claims
`holds a short-lived user credential`
`connects as CONTROL · sheet 3`

D — Domain services
tender, loads, fleet, documents
`holds a short-lived user credential`
`disputes and billing sit here too`

A — Integration adapters
telematics, ERP, maps, payments
`holds a short-lived user credential`
`one connection per tenant served`

K — Workflow workers
they act for one tenant at a time
`holds a short-lived user credential`
`proposed · no V3 ADR`

a replica opens a connection with exactly one user credential · NATS

THE TENANT OWNS THESE · INSIDE ONE ACCOUNT, IN THIS CELL

J — JetStream streams
the durable facts, replayable
`named inside the tenant account`
`stream design is LB-V3-L3-05`

V — NATS KV projections
read models, fast lookup, watch
`a KV bucket is really a stream`
`so it obeys every account rule`

P — Service Postgres
transactional truth, per service
`tenant-scoped, and cell-resident`
`ownership is LB-V3-L3-07`

E — Evidence documents
proof of delivery, photos, papers
`S3-compatible adapter in the cell`
`proposed · no V3 ADR`

everything written here sits inside that one account, in this cell only

WHAT THE LAB MEASURED · DEMO 03, 73 CHECKS PASSED, 0 FAILED

1 — An account is not a vote
3 JetStream accounts, 1 meta group
`a wall for data, not for availability`
`measured · demo 03, T2`

2 — Export is the only door
one direction, subject only
`read-only, and renamed on arrival`
`measured · demo 03, E`

3 — A consumer follows its stream
it lives where the stream lives
`a far replica pays the long link`
`measured · demo 03, T2`

4 — Placement follows the client
a new stream lands where you are
`unless a cluster is named for it`
`measured · demo 03, T2 and T3`

? — L3-01-O01 · one credential per account, or one replica per account?
the biggest open question this sheet raises
`a shared replica serving two accounts needs two connections`
`a replica per account is simpler, and costs far more`
`open · architecture to confirm · no V3 ADR`

! — A replica borrows a user · it never owns an account
the cell is a place; the account is a who
`a replica that held an account would make the cell a tenant`
`that is the mistake sheet 1 exists to stop`
`directional position as at 2026-09-17`

Read it downward. The cell provides the machinery, a replica borrows one user credential, and everything the replica writes lands inside that user's account. **Built today:** nothing in the top three bands. The bottom band is measured — demo 03, 73 checks passed and 0 failed on 2026-09-17. **Not shown:** subjects, streams and grants, which are `LB-V3-L3-05` ; process packaging and scaling, which are `LB-V3-L3-12` ; claim text and permission grants, which are `LB-V3-L3-09` and `L4` .

SCOPE	**One square, opened.** What runs inside one cell, and whose account owns what it writes.
DIRECTION	**Directional** as at 2026-09-17, except the bottom band. That band is measured.
CELL	The cell owns the machinery. Ingress, NATS, workflow and stores exist **once per cell**.
REPLICA	A replica is a **place-thing**. It runs in a cell. It borrows a user; it never owns an account.
CREDENTIAL	One connection carries **one** user credential. Two accounts means two connections.
DATA	Streams, KV, Postgres and documents are **tenant-owned** and stay inside the cell that holds them.
KV	A KV bucket is really a stream. So it obeys every account rule a stream obeys.
HA	Three NATS servers is a **logical minimum**, not a sizing number. Sizing is L4.
MEASURED	Only the four solid tiles are measured. Everything else is proposed or open.
OPEN	**L3-01-O01** is the one to answer first. It changes how every service is deployed.

LEGEND

- **Cyan boundary** — what the cell owns: a place
- **Solid tile** — measured in the lab, with the run named
- **Cyan dotted line** — the cell hosts what is below it
- **Line label** — names the relationship, never bare "uses"
- **Blue boundary** — service replicas that borrow a user
- **Dashed grey tile** — proposed for V3, not built
- **Blue line** — a NATS connection, opened with one credential
- **Amber boundary** — what the tenant account owns
- **Dashed amber tile** — open, a named owner must confirm
- **Amber dashed line** — a write landing inside an account
- **Purple boundary** — facts measured in the demo 03 lab
- **Amber panel** — a rule that governs the whole sheet
- **Arrow head** — the direction of the relationship

WORDS ON THIS SHEET

cell one region's complete runtime
replica one running copy of a service
account the NATS tenancy boundary
user credential a short-lived NATS identity
claim resolver serves account claims to a server
JetStream NATS durable storage and replay
stream a durable, replayable record of facts
consumer a reader's cursor over a stream
KV NATS key-value, a fast lookup store
projection a read model built from a stream
placement which cluster stores a new stream
meta group the JetStream vote across servers
export, import the only path between accounts
WAF web application firewall
MFE micro-frontend, a page part loaded at run time
ERP enterprise resource planning system
HA high availability
ADR architecture decision record

Traceability: L3-01-019 to L3-01-031, L3-01-034 to L3-01-040 and L3-01-049 (cell-owned runtime, service replicas and the borrowed credential, tenant-owned durable state, measured lab facts, derived concerns). Parent: LB-V3-L2-01, its service planes on sheet 1 and its cell contents on sheet 2. Related: LB-V3-L3-05 Messaging / NATS; LB-V3-L3-08 Multi-Region; LB-V3-L3-09 Security + Identity. Evidence: demos/03-multi-cluster-and-accounts/REPORT.md.

Which Account Does Each Service Connect As

Sheet 2 says a replica borrows a user credential. This sheet says **whose**. Every service named on LB-V3-L2-01 is a row; the four account types from sheet 1 are the columns. One connection sees exactly one account, so a service that needs two accounts needs two connections. **This is a directional document** as at 17 September 2026. Nothing here is built. The service names are carried down from LB-V3-L2-01 and are not restated or extended; subject and grant text belongs to LB-V3-L3-05; permission claims belong to LB-V3-L3-09 and L4.

SERVICES, DOWN · ACCOUNT TYPES, ACROSS — ONE CONNECTION SEES EXACTLY ONE ACCOUNT

services, down / accounts, across	$SYS · administration	CONTROL · control plane	PLATFORM · shared data	TENANT · one per tenant
CONTROL PLANE SERVICES · THEY RUN THE PLATFORM, NOT A TENANT'S BUSINESS				
Tenant + organisation — lifecycle of a tenant and its orgs	not used	Connects — owns tenant and org records	not used	Imports — lifecycle events, read-only
Identity + membership — who a person is, and where they belong	not used	Connects — owns persons and memberships	not used	Open — 007 · control, or per tenant?
NATS provisioning — accounts, credentials, claims, placement	Connects — pushes claims · system API	Connects — reads the placement record	not used	Refused — it never acts as a tenant
Reference + config — the shared corpus every tenant reads	not used	not used	Connects — owns the one shared corpus	Imports — every tenant, read-only
Entitlements + admin — what a tenant has bought and may use	not used	Connects — owns entitlement records	not used	Imports — a tenant reads its own
Observe + audit — health, telemetry and the audit trail	Connects — server, JetStream health	Connects — collector, Grafana, audit	not used	Open — 007 · control, or per tenant?
DOMAIN PLANE SERVICES, ADAPTERS AND WORKERS · THEY ACT FOR ONE TENANT				
Marketplace + tender — offers, bids and award	Refused — domain service, not an admin	not used	Imports — reference data, read-only · 008	Connects — acts as one tenant at a time
Loads + trips + routes — the movement of goods	Refused — domain service, not an admin	not used	Imports — reference data, read-only · 008	Connects — acts as one tenant at a time
Fleet + tracking — vehicles, drivers and position	Refused — domain service, not an admin	not used	Imports — reference data, read-only · 008	Connects — acts as one tenant at a time
Documents — proof of delivery and paperwork	Refused — domain service, not an admin	not used	Imports — document types, read-only	Connects — acts as one tenant at a time
Disputes + acceptance — claims raised against a trip	Refused — domain service, not an admin	not used	Imports — reference data, read-only · 008	Connects — acts as one tenant at a time
Billing + settlement — invoices, rates and payment	Refused — domain service, not an admin	not used	Imports — currency and rate reference	Connects — acts as one tenant at a time
Integration adapters — telematics, ERP, maps, payments	Refused — domain service, not an admin	not used	Imports — partner and map reference	Connects — one connection per tenant
Workflow workers — long-running work, per tenant	Refused — domain service, not an admin	not used	Imports — reference data, read-only · 008	Connects — one connection per tenant

! Only one service ever connects as $SYS

NATS provisioning, and nothing else on this sheet

control-plane services connect as CONTROL; shared data as PLATFORM
a domain service connecting as $SYS is refused, not merely unusual
as at 2026-09-17, CONTROL added 2026-10-19 · no V3 ADR

? L3-01-O08 · import, or a second connection?

a domain service needs tenant data and shared data

one connection sees one account, so it cannot see both
an import is cheaper; a second connection is simpler to reason about
open · architecture to confirm · no V3 ADR

Read a row for a service and a column for an account. Blue means the service opens a NATS connection as that account. Grey means it only sees that account's data through an export and import, with no connection of its own. Red means the combination is refused. **The shape is the finding:** the control plane connects as `CONTROL`, shared data sits in `PLATFORM`, the domain plane is tenant-shaped, and exactly one service touches `$SYS`. **Built today:** nothing on this sheet. **Not shown:** the subject and grant text, which is `LB-V3-L3-05`; permission claims, which are `LB-V3-L3-09` and `L4`.

SCOPE	**One question.** Whose credential does each service hold? Sheet 2 asks it; this sheet answers it.
DIRECTION	**Directional** as at 2026-09-17. Nothing on this sheet is built, and no V3 ADR exists.
SOURCE	Every service name is carried down from **LB-V3-L2-01**. This sheet adds no service of its own.
RULE	**One connection sees one account.** A service that needs two accounts needs two connections.
SYSTEM	Only **NATS provisioning** connects as `$SYS`. It pushes account claims. That is the whole job.
CONTROL	The control plane connects as **CONTROL**. Only reference data stays in **PLATFORM**, for tenants to import.
DOMAIN	The domain plane is **tenant-shaped**. All eight rows have the same answer, which is the point.
IMPORT	An **import** is read-only and renamed on arrival. Demo 03 measured that. It is the only door.
REFUSED	A red square is a **decision**. A domain service must never hold administrative rights.
OPEN	**L3-01-O07** and **L3-01-O08** are raised by this sheet. Both need architecture to confirm.

LEGEND

- **Blue boundary** — a group of service replicas
- **Amber panel** — a rule that governs the whole sheet
- **Red square, "Refused"** — the platform must never allow it
- **Dashed grey tile** — proposed for V3, not built
- **Blue square, "Connects"** — opens a connection as that account
- **"not used"** — no relationship, and none is needed
- **Dashed amber tile** — open, a named owner must confirm
- **Grey square, "Imports"** — sees the data, holds no connection
- **Column header** — an account type carried from sheet 1
- **Red square or panel** — a combination that is refused
- **Amber square, "Open"** — the answer is not decided yet
- **Row label** — a service carried down from LB-V3-L2-01

WORDS ON THIS SHEET

- **account** the NATS tenancy boundary
- **$SYS** the NATS system account
- **control account** runs the control plane
- **platform account** shared data tenants import
- **tenant account** holds one tenant's own records
- **connects as** opens a NATS link with that account's user
- **import** read-only sight of another account's subjects
- **export** the grant that makes an import possible
- **claim** the signed statement that defines an account
- **system API** the subjects only $SYS may speak on
- **placement record** names a tenant's permitted cells
- **control plane** services that run the platform itself
- **domain plane** services that run a tenant's business
- **ERP** enterprise resource planning system
- **ADR** architecture decision record

Traceability: L3-01-041 to L3-01-050 (service-to-account mapping, the single system-account holder, the control plane in CONTROL, shared data in PLATFORM, the tenant-shaped domain plane, refusals, the import-only reference path). Parent: LB-V3-L2-01, its control plane and domain service plane on sheet 1. Related: LB-V3-L3-05 Messaging / NATS, for export, import and subject text; LB-V3-L3-09 Security + Identity, for permission claims.